Wake Forest University
School of Medicine

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Changes of White Matter Diffusivity After a Season of High School Varsity Football

Alessandra E. Martinez*[1,2], Megan E. Lipford[1], Jeongchul Kim[1], Christopher T. Whitlow[1]

[1]Wake Forest School of Medicine, Winston-Salem, NC

[2]Universidad de Monterrey, Monterrey, N.L., Mx.

Background/Objective

Football has the highest concussion rate among contact sports.

Subconcussions are cranial impacts that don't result in clinically diagnosed concussions. → **effects** are still **unknown**[1]

Ages 10-20 is a critical time for brain development[2].

Retrieved from https://www.southcountyhealth.org/programs-services/orthopedics-center/sports-medicine

Objective

Analyze longitudinal diffusion Magnetic resonance imaging (MRI) metrics and cumulative head impact exposure data.

[1]. Davenport, E. Journal of NeuroTrauma 31:1617–1624. [2]. Paus, T. VOLUME 9, ISSUE 2, P60-68, FEBRUARY 2005

Method: ITAKL Dataset

Demographics of subjects

Variable	Group	
	Football players*	Noncontact sports athletes
Sample size	52	21
Age mean$(\pm SD)$	15.8 ± 1.3	16.4 ± 1.6
Days between scans mean $(\pm SD)$	142.3 ± 51.8	

*No concussion history 6 months prior and during the season

Biomechanical exposure data

Head Impact Telemetry System (HITS™)[3]

RWE$_{CP}$
Risk-weighted exposure to the combined probability of linear and rotational acceleration

$$= \sum_{i=1}^{n_{hits}} \mathbf{CP}(\mathbf{a_L}, \mathbf{a_R})_i$$

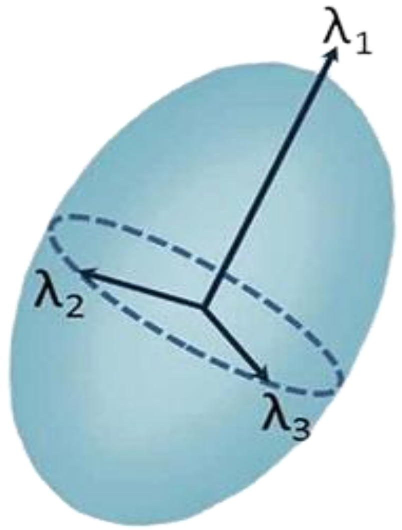
[3] a_L → measured peak linear acceleration
a_R → measured peak rotational acceleration
n_{hits} → number of head impacts in a season for a given player

[3]. Urban, J et al. Annals of Biomedical Engineering 2013 41: 2474–2487.
Figure retrieved from https://press.rsna.org/timssnet/media/pressreleases/14_pr_target.cfm?ID=1905

Method: Diffusion Tensor Imaging

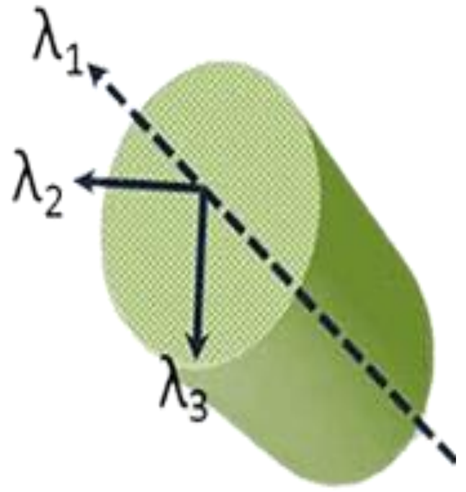
For all subjects:
Pre- and post-season imaging data sets

Fractional Anisotropy [FA]



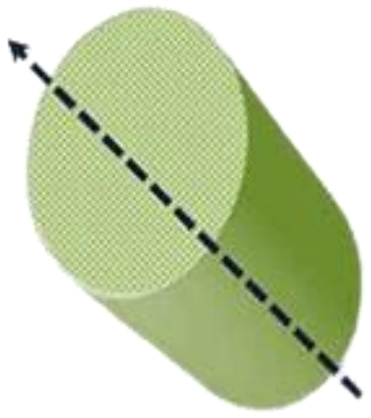
Degree to which diffusion is anisotropic (linearity) [4].

Mean Diffusivity [MD]



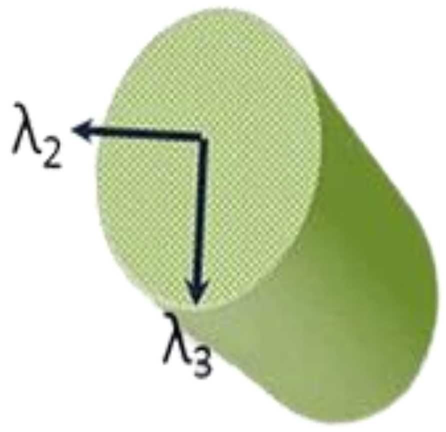
Characterizes the net degree of displacement of the water molecules[4].

Axial Diffusion [AD]



Rate of diffusion in the principal direction[4]. λ_1

Radial Diffusivity [RD]



Rate of diffusion perpendicular to the principal diffusion direction[4].

[4]. J Dev Behav Pediatr. 2010 May; 31(4): 346–356. 3
Figure retrieved from DeSouza, D. October 2016Frontiers in Neuroanatomy 10

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Methods: Analysis

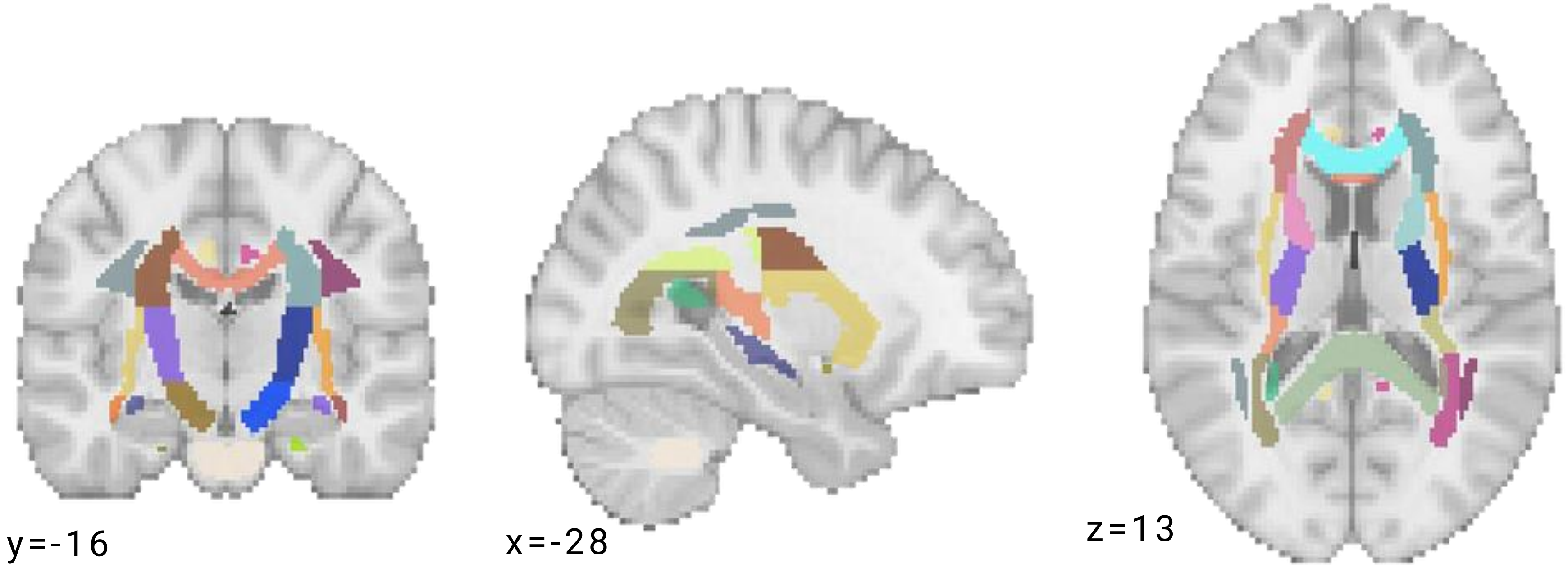


Figure 1. John Hopkins University White Matter (WM) tractography atlas

Two-tail paired t-tests on each WM region of interest (ROI)

DTI metrics

λ_1, λ_2, and λ_3

For ROIs with statistically significant changes in FA and MD post- and pre-season ($p_{FDR} < 0.05$)

Linear regression models

- Association with log-transformed RWEcp
- Correcting for days between scans and age

Results: DTI Metric Changes

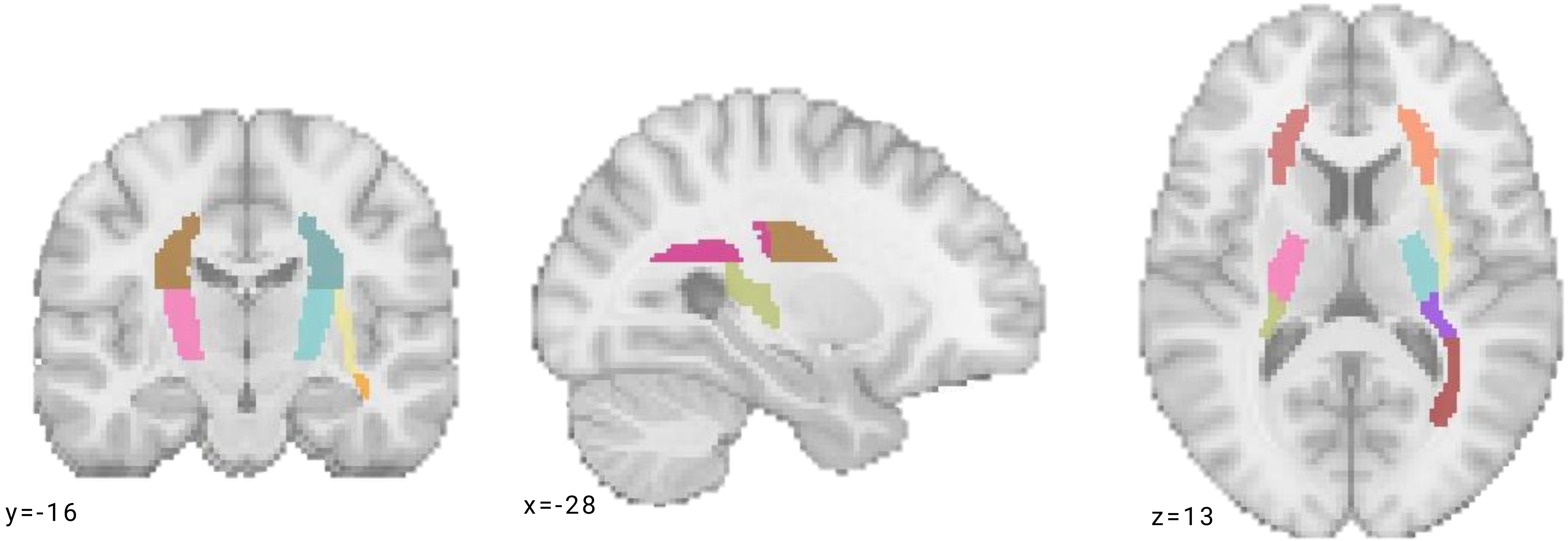


Figure 2. ROIs with significant changes in FA, MD, and RD between pre- and post season

Increase in FA → Increase linearity[5]

Decrease in MD and RD → Decrease in average water diffusion[5]

→ **Dense axonal packing[5]**

[5]. Shenton, M.E., et al, Brain Imaging Behav. 2012 6(2): 137–192

λ_2 and λ_3 to better understand FA and MD Changes

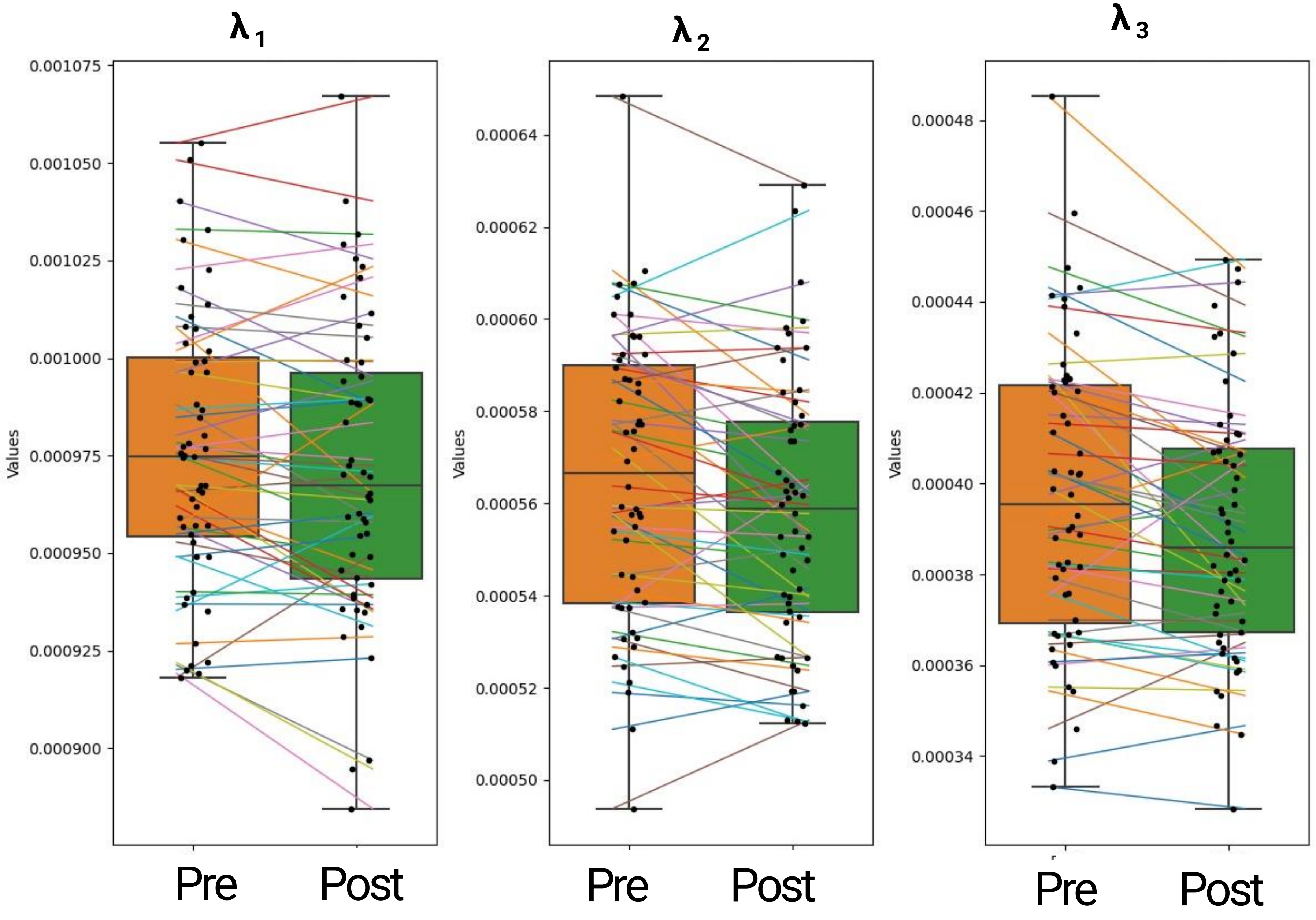
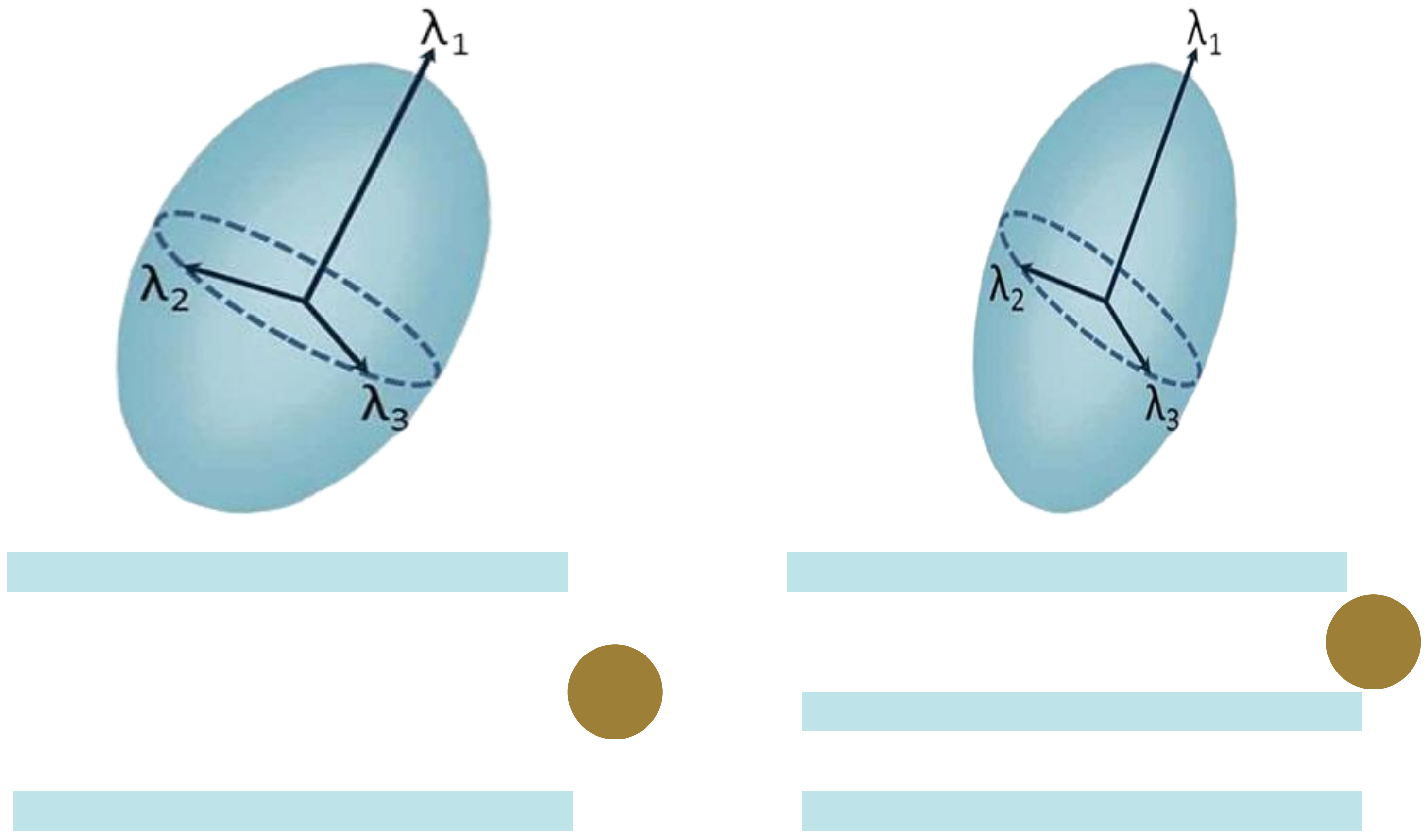


Figure 3. λ_1 , λ_2 ,and λ_3 of Anterior Corona Radiata R

FA and **MD** changes were mainly due to **decreases** $\boldsymbol{\lambda_2}$ and $\boldsymbol{\lambda_3}$



Associated with fiber myelination[6]

White matter maturation aging process[6]

[6]. Provenzale,J. AJR 2012; 198:1403–1408

Correlation with RWE_{cp}

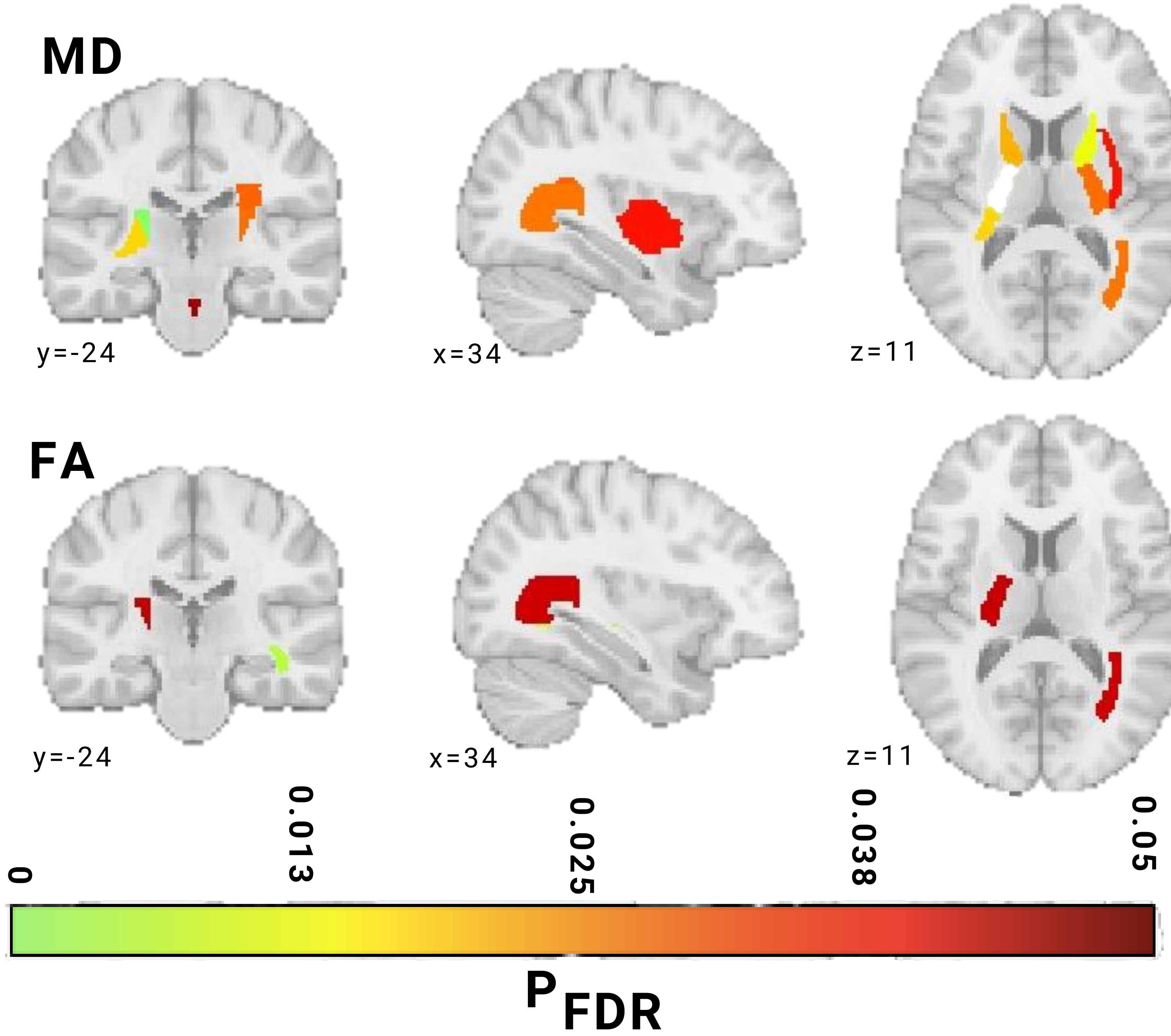
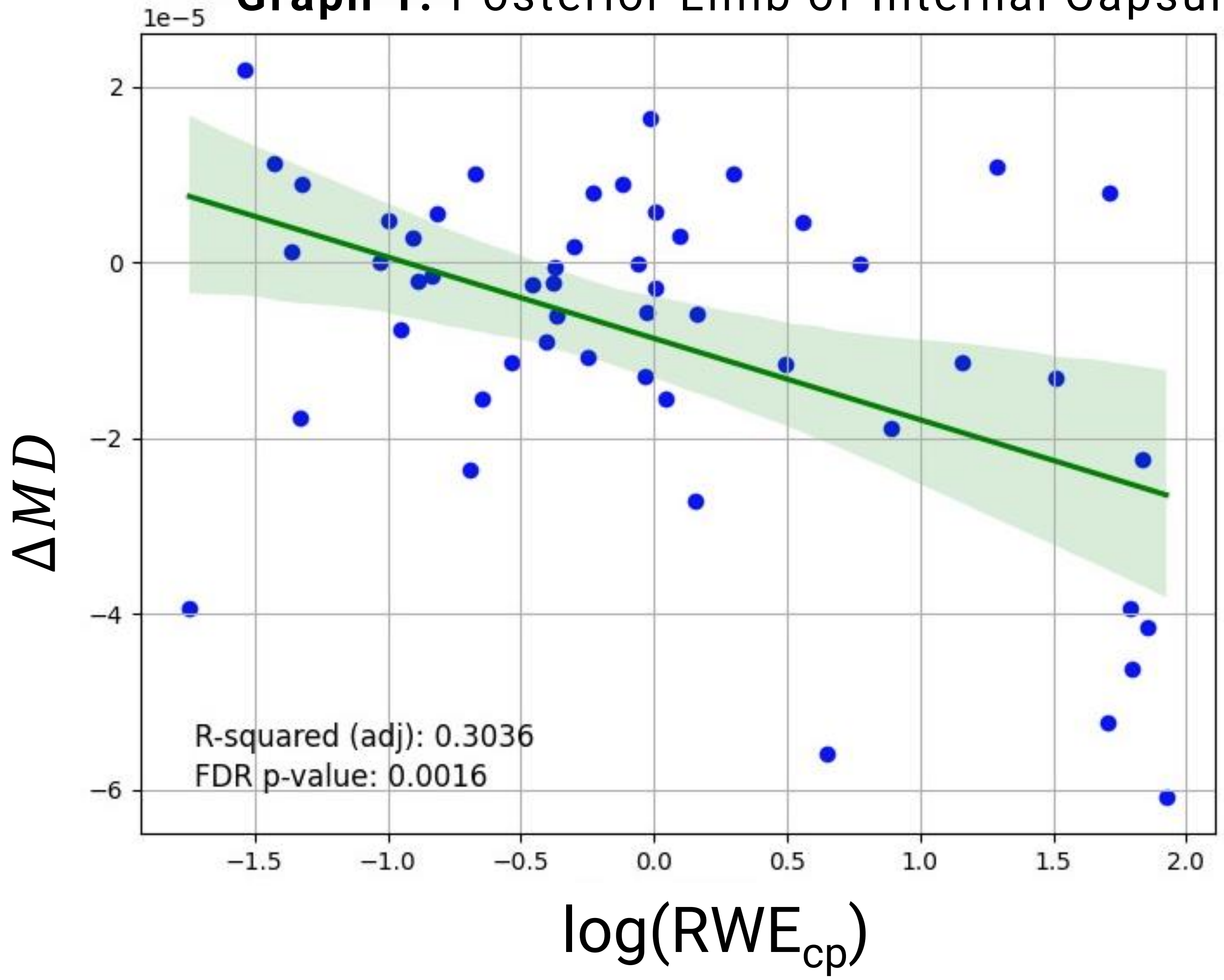


Figure 4. ROIs with significant correlation with Z-scored RWE_{cp}

Graph 1. Posterior Limb of Internal Capsule L



ΔMD was negatively correlated to RWE_{cp}
ΔFA was positively correlated

Coping mechanism → increase in myelination[7]

[7]. Mayinger, M. Brain Imaging and Behavior volume 12, pages44–53 (2018)

Conclusions

- In a season of high school football, there are DTI metric changes
  - **Increases** in **FA** and **decreases** in **MD** and **RD**.
- **FA** and **MD variations** were **mainly due** to **decreases** in $\boldsymbol{\lambda_2}$ and $\boldsymbol{\lambda_3}$
  - Suggests **swelling** and **myelination**.
- In **some ROIs MD** differences had a **significant relationship** with $\mathbf{RWE_{cp}}$
- Results should be carefully interpreted
  - **Limitations don't allow** us to **determine** if these **changes** are **abnormal** to white matter maturation in adolescents.
- **Future research**
  - Voxelwise analysis
  - Second time point for control group

Retrieved from https://www.axios.com/2022/01/07/nfl-wonderlic-test-draft-eliminated

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